

'islands' during manufacture. The islands can alter the wavelength of light—an unacceptable flaw in laser.

The structure of a nitride cavity comprising GaN bulk layer, multiple quantum wells (MQWs) active region consisting of several periods of GaN barrier layer and InGaN active layer in nanometre thickness is shown in Fig. 3(a). On both sides of the nitride cavity, distributed Bragg reflector (DBR) is deposited.

A diode laser is created by depositing layers of semiconductor materials on top of an underlying substrate like a thin wafer of sapphire or gallium nitride. The substrate is placed inside a reaction chamber where hot gases of metal organic vapours deposit successive layers of gallium, indium, nitrogen and other atoms on that substrate, with the exact proportions of each element varying from layer to layer. The atoms in these layers automatically align with the existing crystalline structure, as predetermined by the substrate. A substrate is a slice of crystal and anything grown on top of it inherits its crystalline structure.

On either side of the InGaIn active layer is a GaN barrier layer. GaN layer acting as an optical waveguide above the substrate is doped with silicon impurities to produce an excess of electrons. On the other end, GaN is doped with magnesium to give it an excess of 'holes' or positive charges. It also acts as an optical waveguide.

The total optical thickness of the cavity is about 23.5λ , where ' λ ' is the designed wavelength.

In LEDs, the photons leave the well almost immediately, perhaps rebounding once or twice before exiting the device or being absorbed in other layers. But in laser diodes, which produce coherent light, the photons stay largely confined within the trench. Two highly reflective mirrors—generally polished crystal surfaces at either end of it—recycle the photons back and forth inside, further stimulating electron-hole recombination. The laser light gen-

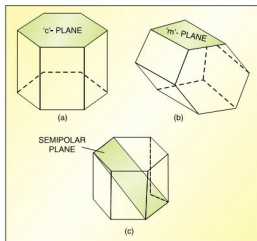


Fig. 4: Crystal plane of substrate

erated by this 'stimulated emission' process is a tight pencil beam of exceedingly pure colour.

Initially, the scientists tried to produce green lasers by depositing layers of semiconducting materials parallel to sapphire substrate's 'c'-plane, which is perpendicular to the crystal's axis of hexagonal symmetry (Fig. 4(a)). Unfortunately, electrostatic forces and internal stresses between successive layers of positively charged gallium or indium ions and negatively charged nitrogen ions create strong electric fields perpendicular to the 'c'-plane. These fields, which can reach up to 100 volts per micron, counteract the applied external voltage. They pull electrons away from holes, making it harder for them to recombine and produce light. As a result, the electrons pile up at one side of the central active layer and holes at the other end, both reluctant to cross over and recombine. The problem worsens as the wavelength shifts towards green from the violet/blue spectrum. This effect is known as quantum-confined Stark effect.

And as the current through the diode increases, the greater number of charge carriers partially blocks the internal electric fields that keep electrons and holes apart. With these fields partially screened out, electrons and holes then recombine at higher energies, shifting the light towards the blue end of the spectrum.

These problems are the main reason why green laser diodes and high-efficiency green LEDs could not be made.

Solution to green gap problem

To find a solution to this problem, scientists replaced the sapphire substrate with a thin wafer of pure, crystalline GaN that was sliced along a larger crystal's 'm'-plane as shown in Fig. 4(b) and then polished. Diodes fabricated on these non-polar substrates

do not encounter the problems of conventional polar 'c'-plane devices, because the troublesome fields caused by polarisation and internal stresses are much lower. Diodes grown on GaN also produce light more efficiently than one grown on sapphire because they suffer from far fewer crystalline defects, submicroscopic irregularities and mismatches at the interfaces between successive layers. Such defects act as centres where electrons and holes recombine to produce unwanted heat instead of light. They can easily propagate upward through the successive diode layers during the growth process and reach the active layers. Diodes grown on non-polar GaN substrate can therefore produce much more light and less heat to dispose of.

Another option to obtain green lasers is use of semipolar substrates (Fig. 4(c)) that are cut at 45° angle to the crystal's major axis. These substrates also do not produce strong fields and they seem to yield better lasers and LEDs than an 'm'-plane substrate.

Researchers hope that 'm' plane and semipolar geometries can be adopted for commercial production of green diodes and high-power green LEDs. However, semiconductors must be grown with extreme care.

Layers of InGaIn are grown at substantially lower temperature—about 700°C versus 1000°C for the GaN lay-